



Ixigo

Coforge
Graduate Engineer Trainee



Content

1:- What, Why, When?

2:- Manual Testing

2.1:- Test Plan

2.2:- Test Scenarios

2.3:- Test Cases

2.4:- Defect Report

2.5:- Requirement Traceability Matrix

3:- Automation Testing

3.1:- Tools and Setup

3.2:- Folder Structure

3.3:- Code Files

3.4:- Allure Reports

3.5:- Cucumber

4:- API Testing

4.1:- Postman

4.2:- Jmeter Report

4.4:- Rest - assured

5:- JDBC

6/22/2026



What?

Testing is the process of verifying and validating the Ixigo application to ensure:

- Correct search results.
- Accurate Pricing
- Proper Booking Flow

Why?

- To prevent booking failures
- To ensure correct fare and availability
- To improve user experience
- To detect bugs early

When?

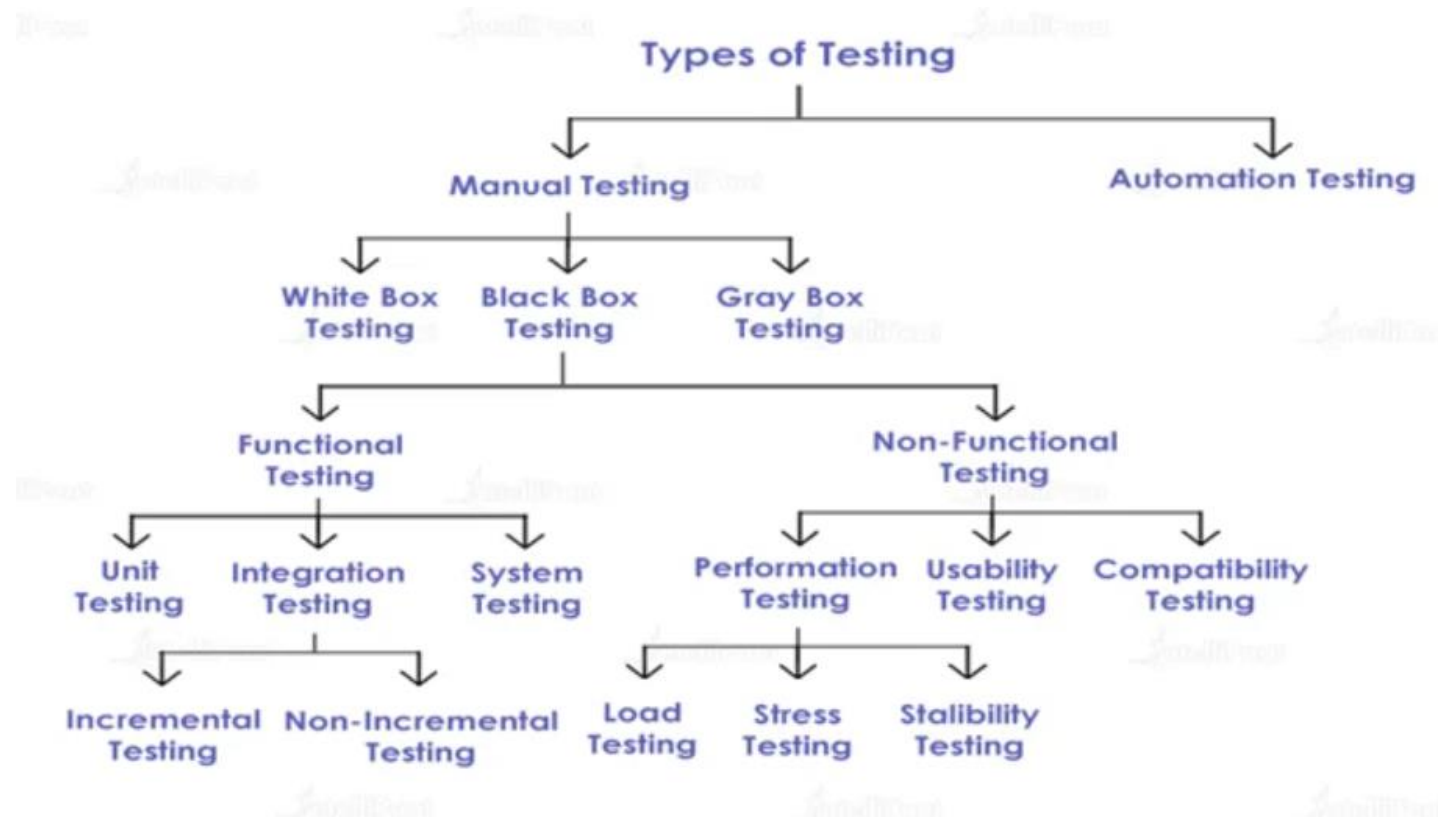
Testing is performed across SDLC :

- Requirement Phase
- Development Phase
- Deployment Phase
- Maintenance Phase

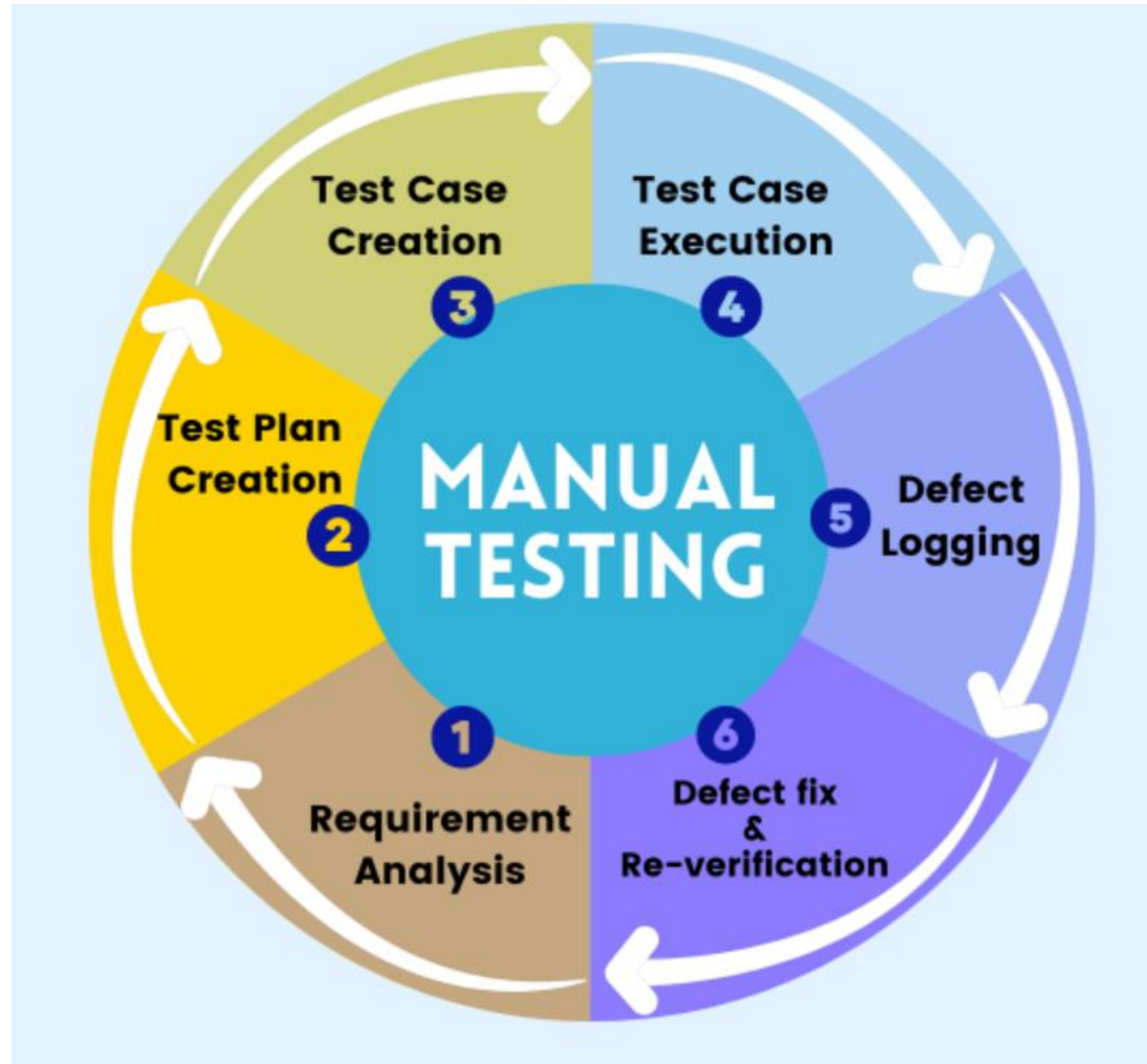


Manual Testing

- Manual testing is performed without Automation tools.
- Testers execute test cases manually to find defects.
- The tester validates the application from an end-user perspective.
- Focuses on functionality, usability, Navigation, and user experience.



Life Cycle Of Manual Testing



Test Plan

Test Plan for Ixigo.com Website

1. Test Plan Identifier

File Name: 1.0_Ixigo_tplan_v1.0.doc

- Document Name: Test Plan – Travel Booking Application
- Version: IX_TR_1.0
- Date: 27-May-2026
- Location: Coforge Ltd, Hyderabad

2. Document History

REV	Author	Edits	Date
1.0	Sahil Kumar Singh, Graduate Engineer Trainee	Initial Draft	27/05/2026

3. Introduction

Launched in 2007, Ixigo is a technology company focused on empowering Indian travellers to plan, book and manage their trips across rail, air, buses and hotels. They assist travellers in making smarter travel decisions by leveraging artificial intelligence, machine learning, and data science led innovations on their OTA platforms, comprising their websites and mobile applications. Their vision is to become the most customer-centric travel company, by offering the best customer experience to their users. They focus on travel utility and customer experience for travellers in the 'next billion user' segment is driven by technology, cost-efficiency and their culture of innovation has made them India's leading travel ecosystem for the 'next billion users'.

4. Test Items

Specifies what components/modules will be tested.

Example:

- User Registration Module
- User Login Module
- Flight/Train Search Module
- Booking Management Module
- Payment Gateway Module
- Offer & Discount Module
- Profile Management Module
- Notification Module
- Logout Functionality

5. Features to be Tested

Medium priority features are features that have been stable and High Priority features are features that have been recently changed.

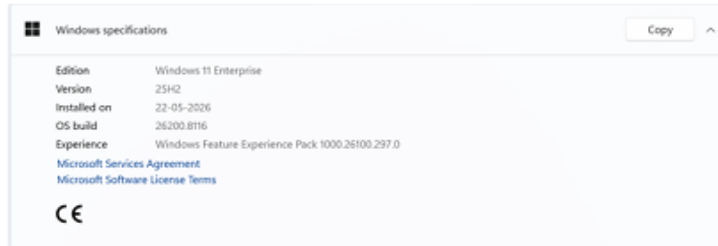
Feature to be Tested	Priority	Comments
User Registration	High	
User Login	High	
Flight Search	High	
Train Search	High	
Bus Search	High	
Hotel Booking	High	
Offers & Coupons	High	
Payment	High	
Logout	High	

6. Features NOT to be Tested

- Help Contents and user manuals will not be tested.
- Installation and presentation will not be tested.
- Support for the other architectures also will not be tested.
- Third-party payment gateway internal logic will not be tested.
- Legacy modules not impacted by current release
- Experimental or beta features will not be tested

7. Test Environment

LP-AZ-vf9ficydR HP ProBook 440 G7		Rename this PC
Device specifications		Copy ^
Device name	LP-AZ-vf9ficydR	
Full device name	LP-AZ-vf9ficydR\IN.COFORGETECH.COM	
Processor	Intel(R) Core(TM) i7-10510U CPU @ 1.80GHz (2.30 GHz)	
Installed RAM	16.0 GB (15.8 GB usable)	
Device ID	858AD706-222D-4983-856D-25FEA7DEBACA	
Product ID	00329-00000-00003-AA503	
System type	64-bit operating system, x64-based processor	
Pen and touch	No pen or touch input is available for this display	



8. Test Approach / Test Strategy

Describes **how testing will be performed**.

The test strategy for Ixigo1.0 will include manual testing with automated testing(**Selenium With Java**) where time/need permits. Automated testing will be used to test the basic Interface objects keyboard and mouse navigation or any redundant tasks best translated to automation. Automated testing will be limited to a Win-64 platform because of Automation tool limitations.

- Sanity test will include an very short battery of tests to ensure the Ixigo1.0 runs/launches. Test execution will happen as individual component testing. These tests will be fully automated.
- Acceptance tests** will include a short battery of tests that will only touch major features to ensure Ixigo1.0 will operate smoothly at a basic level. These tests will be executed on **Windows 11 Enterprise**. The goal will be to automate all of these tests.
- Full Functional tests will include a **comprehensive test** of all features listed in this document. These tests will be limited to the Windows 11 pro. These test will include automation with time allowing.

8.1 Types of Testing

- Smoke Testing
- Sanity Testing
- Functional Testing
- Integration Testing
- System Testing
- Regression Testing

8.2 Test Levels

- Unit Testing (by developers)
- Integration Testing
- System Testing
- Acceptance Testing

8.3 Test Techniques

- Black Box Testing
- Boundary Value Analysis
- Equivalence Partitioning
- Error Guessing **Functionality Testing**

Usability Testing

The usability testing will be accomplished by verifying the information in each window are accurate. Menus, icons and toolbar functionality will be tested as applicable to the navigation and results panes. Multi Window Overlapping will be tested because product supports opening of multiple documents.

Robustness/Reliability Testing

Test the GUI for correct keyboard and mouse navigation, windows, menus, buttons, etc. These tests will include Keyboard only navigation of Ixigo1.0, Mouse only navigation. Correctness and existence of warning/error dialog boxes. The correct/expected functionality of buttons, icons and menus.

Stress Testing

Not applicable

9. Entry and Exit Criteria

9.1 Entry Criteria

Conditions to start testing:

- Requirements approved
- Test environment ready
- Test data available
- Build deployed

9.2 Exit Criteria

Conditions to stop testing:

- All planned test cases executed
- Critical and high defects closed
- Test summary report prepared

10. Test Deliverables

Test plan	1.0_Ixigo_tplan.doc
Test Specification	1.0_Ixigo_tspec.doc
Test Automation Plan	TBD
Test Checklist	TBD
Test Logs	TBD
Test tools	Selenium 4.38, Postman, Rest Assured, JMeter, Jenkins, Git and Git Hub and Appium.

11. Testing Tasks & Resource Needs

- Write Test Specification.
- Write Test Automation plan.
- Write Test Checklist.
- Setup workspace.
- Develop Automation script.

➤ Minimum requirements for platforms:

12. Schedule

Test Documentation

Document	Development Time		Sign-off Date	Dependencies
	Person-Days	Completion Date		
Test Plan	1day	27/05/2026	27/05/2026	Functional Spec. Review
Test Specifications	1 day	27/05/2026	27/05/2026	Test plan, Functional Spec, Review
Test Logs	TBD	27/05/2026		
Test Summary Report	TBD	27/05/2026		
Checklist	TBD	27/05/2026		

Test Development

<i>Deliverable</i>	<i>Development Time</i>		<i>Dependencies</i>
	<i>Person-Days</i>	<i>Completion Date</i>	
Functional Test Suite	1	27/05/2026	
Automation build			

*Approval required				
Project Manager				
Project Manager			Raghavendra BN	Raghavendra BN
Client Approval	Coforge			

<i>Test Execution</i>					
<i>Test Activity</i>		<i>Execution Time (Person-Days)</i>			<i>Dependencies</i>
		Alpha	Beta	FCS	
Acceptance		1 Day	1 Day	1Days	
Functional	Functionality	1	1		
	Usability	All High to Medium test cases			
	Robustness				
	Stress				
	Multi Inst.				
Regression					
Documentation Review		1	1	1	
Total Test Cycle		1	1	1	

13. Risks and Assumptions

Risks:

- **Environment Instability:** Delays in setting up the test environment could impact the testing execution.
- **Resource Availability:** Limited resource availability may impact schedules.
- **Late Requirement Changes:** requirement changes during testing may increase rework.
- Payment gateway downtime may interrupt transaction testing.

Contingencies:

- **Backup Resources:** Have backup resources ready to join the testing effort if required.
- **Flexible Schedule:** Allocate buffer time in the schedule to accommodate unexpected delays.
- **Regular Communication:** Hold daily stand-up meetings to quickly address any issues that arise and resolve blockers quickly.

14. Approvals \Sign of

POSITION	NAME	SIGNATURE	REVIEWED	APPROVAL
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Test Scenarios

Name: Sahil Kumar Singh
Email ID: sahilsinghrnc@gmail.com
Emp ID: 85009884

TEST SCENARIOS FOR IXIGO.COM

Scenario ID	Scenario Name	Scenario Description	Preconditions	Expected Outcome
TS_01	Verify user registration	Validate that a new user can register using valid details	User is not registered	User account is created successfully
TS_02	Verify user login	Validate that a registered user can log in with valid credentials	User must be registered	User is logged in and redirected to dashboard
TS_03	Flight Search	Verify that user is able to search for available flights successfully	Requires search fields should not be empty	Available flights matching the entered search details should be displayed successfully
TS_04	Train Search	Verify that user is able to search for available train successfully	Requires search fields should not be empty	Available trains matching the entered search details should be displayed successfully
TS_05	Bus Search	Verify that user is able to search for available bus successfully	Requires search fields should not be empty	Available buses matching the entered search details should be displayed successfully
TS_06	Hotel Booking	Verify that the user is able to search and book a hotel successfully by entering valid details	User should enter a valid city and state	User should be able to search and book a hotel successfully
TS_07	Offer & Discount	Verify that the user is able to view and apply available codes, or discounts successfully during booking	User should proceed to the payment or checkout page	Eligible offers and discounts should be displayed correctly to the user

Test Cases

Project Name:	Ixigo.com						
Module Name:	End to End test	Date Updated:	27/05/2026				
Last Build Tested:		Reviewed By:	Raghavendra BN				
Run No:	8 Test Cases	Executed By:	Sahil Kumar Singh				
Run Date:	27/05/2026						
Test Objectives:	Mapping Test Scenarios with Test Cases (Ixigo.com)						
Test Setup:	MS Excel Sheet (Online)						
Req_Id	Test_Case ID	Test_Case_Description	Test_steps	Test_Data	Expected_Result	Actual_Result	Status
Req_1.0	TC_01	Validate that a new user can register using valid details	1.User should open Microsoft Edge browser 2. User should navigate to https://www.ixigo.com/ 3. User should click on Register option on the Ixigo homepage. 4. User should enter valid Full Name, Email ID, Password and Mobile Number 5. User should enter City details 6. User should click on Submit / Save button	Url: https://www.ixigo.com/ Name: Sahil Email: sahil.snp@gmail.com Password: ***** Mobile: +91 9853745986 City: Ranchi	User account should be created successfully and user should be redirected to OTP page	User account created successfully	Passed
Req_1.1	TC_02	Verify user can login with valid credentials	1.User should open Microsoft Edge browser 2. User should navigate to https://www.ixigo.com/ 3. User should click on Login option on the Ixigo Homepage. 4. User should enter valid Email ID and Password or enter valid mobile number and enter the OTP.	Url: https://www.ixigo.com/ Email: sahil.snp@gmail.com Password: *****	User should be logged in successfully and Dashboard page should be displayed	User logged in successfully and Dashboard page should be displayed	Passed

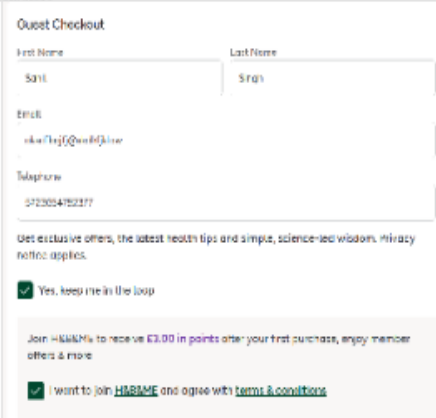
Req_1.2	TC_03	Verify that user is able to search for available flights successfully	1. User should open Microsoft Edge browser 2. User should navigate to https://www.ixigo.com/ 3. User should click on Login option on the ixigo Homepage. 4. User should enter valid Email ID and Password or valid Mobile Number and OTP. 5. User should click on Login button 6. User should observe the	Url: https://www.ixigo.com/ Email: sahil.snp@gmail.com Password: ***** From: Delhi To: Mumbai Departure Date: **** Travellers: 2 Adults Class: Economy	Available flights should be displayed successfully based on the search criteria	Available flights displayed successfully	Passed
Req_1.3	TC_04	Verify that user is able to search for available train successfully	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.ixigo.com/ 3. User should click on Trains tab. 4. User should enter From station and To station. 5. User should select Journey date. 6. User should select travelclass if required. 7. User should click on Search button.	Url: https://www.ixigo.com/ Email: sahil.snp@gmail.com Password: ***** From: Delhi To: Mumbai Departure Date: **** Travellers: 2 Adults Class: 3AC	Available trains should be displayed successfully based on the search criteria	Available trains displayed successfully	Passed

Req_1.4	TC_05	Verify user is able to search for available bus successfully	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.ixigo.com/ 3. User should click on Buses tab. 4. User should enter Source city and Destination city. 5. User should select Travel date. 6. User should click on Search button.	Url: https://www.ixigo.com/ From: Hyderabad To: Bangalore Travel Date: ****	Available buses should be displayed successfully	Available buses displayed successfully	Passed
Req_1.5	TC_06	Verify that the user is able to search and book a hotel successfully by entering valid details	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.ixigo.com / 3. User should click on Hotels tab. 4. User should enter Destination city/location. 5. User should select Check-in and Check-out dates. 6. User should select number of guests and rooms. 7. User should click on Search button. 8. User should select a hotel from the available list. 9. User should click on Book Now button. 10. User should enter guest details and payment details. 11. User should click on Confirm Booking button.	Url: https://www.ixigo.com/ Location: Goa Check-In Date: ***** Check-Out Date:***** Guest: 2 Adults Payment Method: Credit Card/UPI	Hotel booking should be completed successfully booking confirmation should be displayed.	Hotel booked successfully and confirmation displayed	Passed

Req_1.6	TC_07	Verify that the user is able to cancel an existing booking successfully from the booking section	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.ixigo.com / 3. User should log in with valid credentials. 4. User should navigate to My Trips/Bookings section. 5. User should select an existing booking from the booking list. 6. User should click on Cancel Booking option. 7. User should select the reason for cancellation if required. 8. User should click on Confirm Cancellation button.	Url: https://www.ixigo.com / Existing Booking ID: *****	Selected booking should be cancelled successfully and cancellation confirmation should be displayed	Booking cancelled successfully	Passed
Req_1.7	TC_08	Verify that the user is able to log out successfully from the application	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.ixigo.com / 3. User should log in with valid credentials. 4. User should click on Profile/My Account option. 5. User should click on Logout button.	Url: https://www.ixigo.com / Valid Login Credentials	User should be logged out successfully and redirected to login/home page	User logged out successfully	Passed

Defect Report

Defect ID	Defect Description	Steps to Reproduce	Priority	Severity	Status	Created By	Assigned To	Screenshot
1	Cart icon count does not update after removing item from cart.	1. Launch Microsoft Edge Browser. 2. Launch the Holland & Barrett application/website. 3. Log in with valid user account. 4. Navigate to any product listing page or product detail page. 5. Add multiple products to the cart(e.g., 5 products). 6. Open the cart page. 7. Remove one product from the cart using the "-" button. 8. Wait for the cart to refresh/update. 9. Observe the item count displayed on the cart icon in the header/navigation bar.	High	Critical	New	Sahil Kumar Singh	Raghavendra BN	
2	Product priced at exactly \$15.00 is displayed in the "Deals under \$15" section.	1. Launch Microsoft Edge Browser. 2. Open Holland & Barrett website. 3. Navigate to the Offers/Deals section on HomePage. 4. Open the "Deals Under \$15" category. 5. Browse the products listed in the section. 6. Observe that a product with a price of exactly \$15.00 is displayed.	Medium	Minor	New	Sahil Kumar Singh	Raghavendra BN	

3	Search results display irrelevant products.	1. Launch Microsoft Edge Browser. 2. Open the Holland & Barrett website. 3. Navigate to the search bar. 4. Enter "Vitamin C" in the search field. 5. Click search or press Enter. 6. Observe the search results displayed.	High	Medium	New	Sahil Kumar Singh	Raghavendra BN	
4	Search functionality displays results for invalid or random keywords.	1. Launch Microsoft Edge Browser. 2. Open Holland & Barrett website. 3. Enter random text like "wjegfhqwxhu" in the search bar. 4. Click on the search icon. 5. Observe the search results displayed.	Medium	Major	New	Sahil Kumar Singh	Raghavendra BN	
5	Phone number and Email field accepts invalid input during guest checkout without validation	1. Launch Microsoft Edge Browser. 2. Open the Holland & Barrett website. 3. Add any product to cart. 4. Navigate to checkout page. 5. Enter valid details in all fields except phone number and email. 6. Enter invalid phone number (e.g., 186723553 or random digits). 7. Click on Continue as a guest. 8. Observe system behaviour.	High	Major	New	Sahil Kumar Singh	Raghavendra BN	

Requirement Traceability Matrix

Req No.	Requirement Description	Test Case ID	Status
R_1.0	The system should allow a new user of Ixigo.com Registration Module to register successfully using valid name, email ID, mobile number and password details.	TC_01	TC_01 – Passed
R_1.1	The system should allow a registered user of Ixigo.com Login functionality Module to log in successfully using valid credentials	TC_02	TC_02 – Passed
R_1.2	The system should allow the user to search for available flights in Ixigo.com Flight Search Module using valid travel details.	TC_03	TC_03 – Passed
R_1.3	The system should allow the user to search for available trains in Ixigo.com Train Search Module using valid source station, destination station, journey fate, travel class details..	TC_04	TC_04 – Passed

R_1.4	The system should allow the user to search for available buses in Ixigo.com Bus Search Module using valid source city, destination city, and travel date details.	TC_05	TC_05 – Passed
R_1.5	The system should allow the user to search and book hotels successfully in Ixigo.com Hotel Booking Module using valid location, check-in/check-out dates, guest details payment information.	TC_06	TC_06 – Passed
R_1.6	The system should allow the user to cancel an existing booking successfully from the Ixigo.com My Trips/Booking Module and display the cancellation confirmation.	TC_07	TC_07 – Passed
R_1.7	The system should allow the user to log out successfully from the application using Ixigo.com Profile/My Account Module.	TC_08	TC_08 – Passed

Peer Review



Peer Review

2 messages

Sahil Singh <sahilsinghrnc@gmail.com>
To: Raghavendra Bn <raghu.astepahead@gmail.com>

Mon, 22 Jun, 2026 at 9:32 am

Hi Sir,

Hope you are doing well.

I have shared the test deliverables for peer review, including the Test Plan, Test Scenarios, Test Cases, Defect Report, and RTM.

Kindly review them and share your feedback or suggestions at your convenience.

Thank you for your time and support.

Regards,
Sahil Kumar Singh

2 attachments

 **Testplan_Ixigo_SahilKumarSingh(85009885) (1).pdf**
321 KB

 **Ixigo_Testing_Documentation_SahilKumarSingh(85009884) 1 (1).xlsx**
482 KB

Raghavendra Bn <raghu.astepahead@gmail.com>
To: Sahil Singh <sahilsinghrnc@gmail.com>

Mon, 22 Jun, 2026 at 10:16 am

Hi Sahil,

Thank you very much for sending across the test deliverables. I have successfully received all the documents, including the Test Plan, Test Scenarios, Test Cases, Defect Report, and the Requirements Traceability Matrix (RTM). I appreciate the effort and attention to detail that has gone into preparing these materials.

I will reviewed each of the documents to ensure that all aspects of the testing process have been adequately covered. My review will focus on the completeness and accuracy of the test cases, the clarity of the test scenarios, and the traceability between requirements and test cases as documented in the RTM. Additionally, I will examine the defect report to understand the issues identified so far and how they have been addressed.

If there are any particular sections or areas within these documents that you would like me to pay special attention to, please let me know. Your input will help me provide more targeted and constructive feedback.

I have completed my review, I will compile my observations and share detailed feedback with you. If I have any questions or require further clarification during the review process, I will reach out accordingly.

Thank you again for your hard work and for keeping me in the loop. Looking forward to collaborating further to ensure the quality and success of this project.

Good To Go

Best regards,
Raghavendra BN

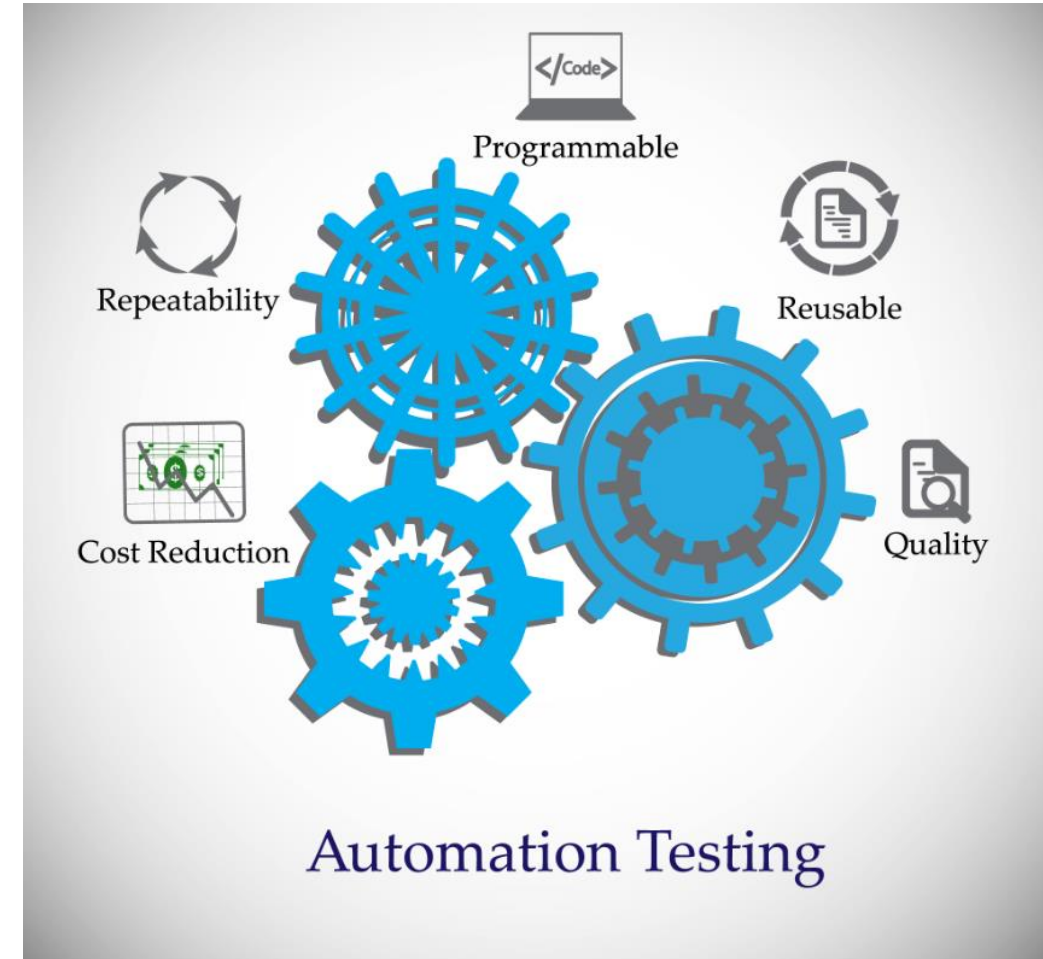
6/22/2026

Automation Testing

Automation testing is a software testing technique where test cases are executed using automation Tools and scripts instead of manual efforts.

It is mainly used to test repetitive and time consuming Scenarios quickly and accurately.

- It saves time and effort.
- Improves accuracy, reduces human errors.
- Useful for regression testing.
- Supports faster execution



Tools & Setup

JDK(Java)

TestNG

PostMan

Allure Reports

MySQL Workbench

JDBC

Selenium

ApachePOI

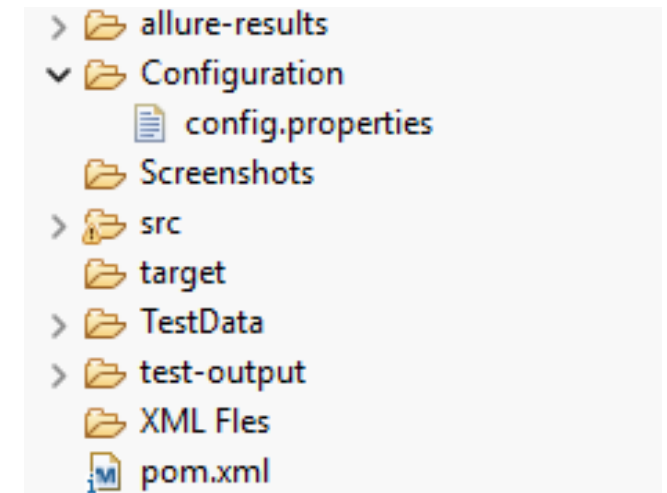
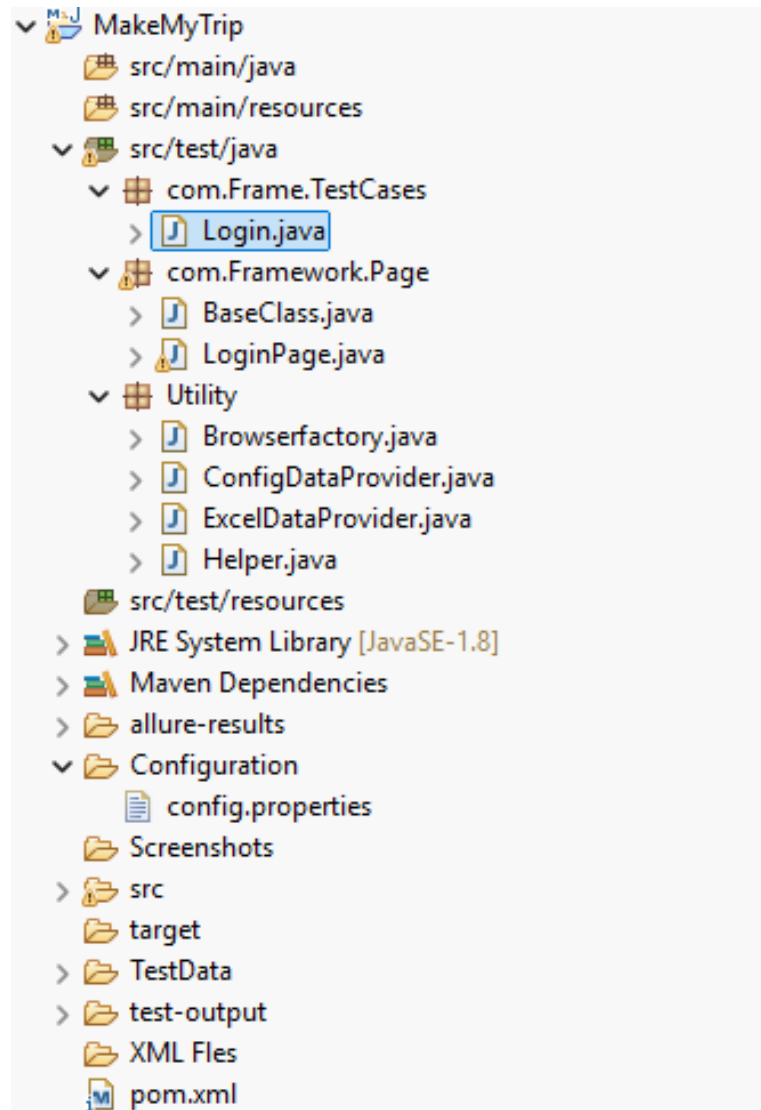
Eclipse

Apache JMeter

Jenkins

Cucumber

Folder Structure



POM.xml

```
<version>0.0.1-SNAPSHOT</version>

<dependencies>
  <!-- Source: https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
  <dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.44.0</version>
    <scope>compile</scope>
  </dependency>
  <!-- Source: https://mvnrepository.com/artifact/org.testng/testng -->
  <dependency>
    <groupId>org.testng</groupId>
    <artifactId>testng</artifactId>
    <version>7.12.0</version>
    <scope>test</scope>
  </dependency>
  <!-- Source: https://mvnrepository.com/artifact/au.com.bytecode/opencsv -->
  <dependency>
    <groupId>au.com.bytecode</groupId>
    <artifactId>opencsv</artifactId>
    <version>2.4</version>
    <scope>compile</scope>
  </dependency>

  <!-- Source: https://mvnrepository.com/artifact/org.apache.poi/poi -->
  <dependency>
    <groupId>org.apache.poi</groupId>
    <artifactId>poi</artifactId>
    <version>5.5.1</version>
    <scope>compile</scope>
  </dependency>
  <!-- Source: https://mvnrepository.com/artifact/org.apache.poi/poi-ooxml -->
  <dependency>
    <groupId>org.apache.poi</groupId>
    <artifactId>poi-ooxml</artifactId>
    <version>5.5.1</version>
    <scope>compile</scope>
  </dependency>
  <!-- Source: https://mvnrepository.com/artifact/io.qameta.allure/allure-testng -->
  <dependency>
    <groupId>io.qameta.allure</groupId>
    <artifactId>allure-testng</artifactId>
    <version>2.35.2</version>
    <scope>compile</scope>
  </dependency>
  <!-- Source: https://mvnrepository.com/artifact/org.apache.maven.plugins/maven-surefire-report-plugin -->
  <dependency>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-surefire-report-plugin</artifactId>
    <version>3.5.6</version>
    <scope>compile</scope>
  </dependency>
</dependencies>
```

Helper.java

```
package Utility;

import java.io.File;

public class Helper {
    public static void capturedScreenShot(WebDriver driver) {

        try {
            TakesScreenshot ts = (TakesScreenshot) driver;
            File source = ts.getScreenshotAs(OutputType.FILE);
            FileUtils.copyFile(source, new File("./Screenshots" + getCurrentDateTime() + ".png"));
            System.out.println("captured ScreenShot - On Failures");
        } catch (Exception e) {
            System.out.println("Exception while Taking ScreenShot" + e.getMessage());
        }

    }

    public static String getCurrentDateTime() {
        DateFormat customFormat = new SimpleDateFormat("MM_dd_yyyy_HH_mm_ss");
        Date currentdate = new Date();
        return customFormat.format(currentdate);
    }
}
```

BrowserFactory.java

```
package Utility;

import org.openqa.selenium.WebDriver;

public class Browserfactory {
    WebDriver driver;

    public static WebDriver BrowserOptions(WebDriver driver, String browserName, String AppUrl) {
        if(browserName.equalsIgnoreCase("chrome")) {
            driver = new ChromeDriver();
        } else if(browserName.equalsIgnoreCase("firefox")) {
            driver = new FirefoxDriver();
        } else if(browserName.equalsIgnoreCase("edge")) {
            driver = new EdgeDriver();
        } else {
            System.out.println("Browser does not support.");
        }

        driver.manage().window().maximize();
        driver.get(AppUrl);
        return driver;
    }

    public static void quitBrowser(WebDriver driver) {
        driver.quit();
    }
}
```


ConfigDataProvider.java

```
package Utility;

import java.io.File;

public class ConfigDataProvider {
    Properties pro;
    public ConfigDataProvider() {
        File src = new File("./Configuration/config.properties");
        try {
            FileInputStream fis = new FileInputStream(src);
            pro = new Properties();
            pro.load(fis);
        } catch (Exception e) {
            System.out.println("Unable to load excel file " + e.getMessage());
        }
    }
    public String getBrowser() {
        return pro.getProperty("Browser");
    }
    public String getAppURL() {
        return pro.getProperty("AppUrl");
    }
}
```

ExcelDataProvider.java

```
package Utility;

import java.io.File;

public class ExcelDataProvider {
    XSSFWorkbook wb;

    public ExcelDataProvider() {
        File src = new File("./TestData/TestData.xlsx");
        try {
            FileInputStream fis = new FileInputStream(src);
            wb = new XSSFWorkbook(fis);
        } catch (Exception e) {
            System.out.println("Unable to load Xls file " + e.getMessage());
        }
    }
    public String getStringData(int sheetIndex, int row, int column) {
        return wb.getSheetAt(sheetIndex).getRow(row).getCell(column).getStringCellValue();
    }
    public String getStringData(String sheetName, int row, int column) {
        return wb.getSheet(sheetName).getRow(row).getCell(column).getStringCellValue();
    }
    public double getNumericData(String sheetName, int row, int column) {
        return wb.getSheet(sheetName).getRow(row).getCell(column).getNumericCellValue();
    }
}
```


BaseClass.java

```
package com.Framework.Page;

import org.openqa.selenium.WebDriver;

public class BaseClass {
    public WebDriver driver;
    public Utility.ExcelDataProvider excel;
    public ConfigDataProvider config;

    @BeforeSuite
    public void Setup() {
        excel = new ExcelDataProvider();
        config = new ConfigDataProvider();
    }

    @BeforeClass
    public void BrowserTest() {
        driver = Browserfactory.BrowserOptions(driver, config.getBrowser(), config.getAppURL());
    }

    @AfterClass
    public void tearDown() {
        Browserfactory.quitBrowser(driver);
    }

    @AfterMethod
    public void tearDownMethod(ITestResult result) {
        if(result.getStatus() == ITestResult.FAILURE) {
            Helper.capturedScreenShot(driver);
        }
    }
}
```

6/22/2026

LoginPage.java

```
package com.Framework.Page;

import org.openqa.selenium.WebDriver;

public class LoginPage {

    WebDriver driver;
    public LoginPage(WebDriver driver) {
        this.driver = driver;
    }
    @FindBy(id = "log_email")
    WebElement email;

    @FindBy(name = "log_password")
    WebElement password;

    @FindBy(xpath = "//input[@name='log_submit']")
    WebElement loginbtn;

    //Create Method
    public void login_EasyCal(String uname, String pass) {
        email.sendKeys(uname);
        password.sendKeys(pass);
        loginbtn.click();
    }
}
```

25

Login.java

```
package com.FrameWork.TestCases;

import org.openqa.selenium.support.PageFactory;
import org.testng.annotations.Test;

import com.FrameWork.Page.BaseClass;
import com.FrameWork.Page.LoginPage;
import com.Utility.ExcelDataProvider;

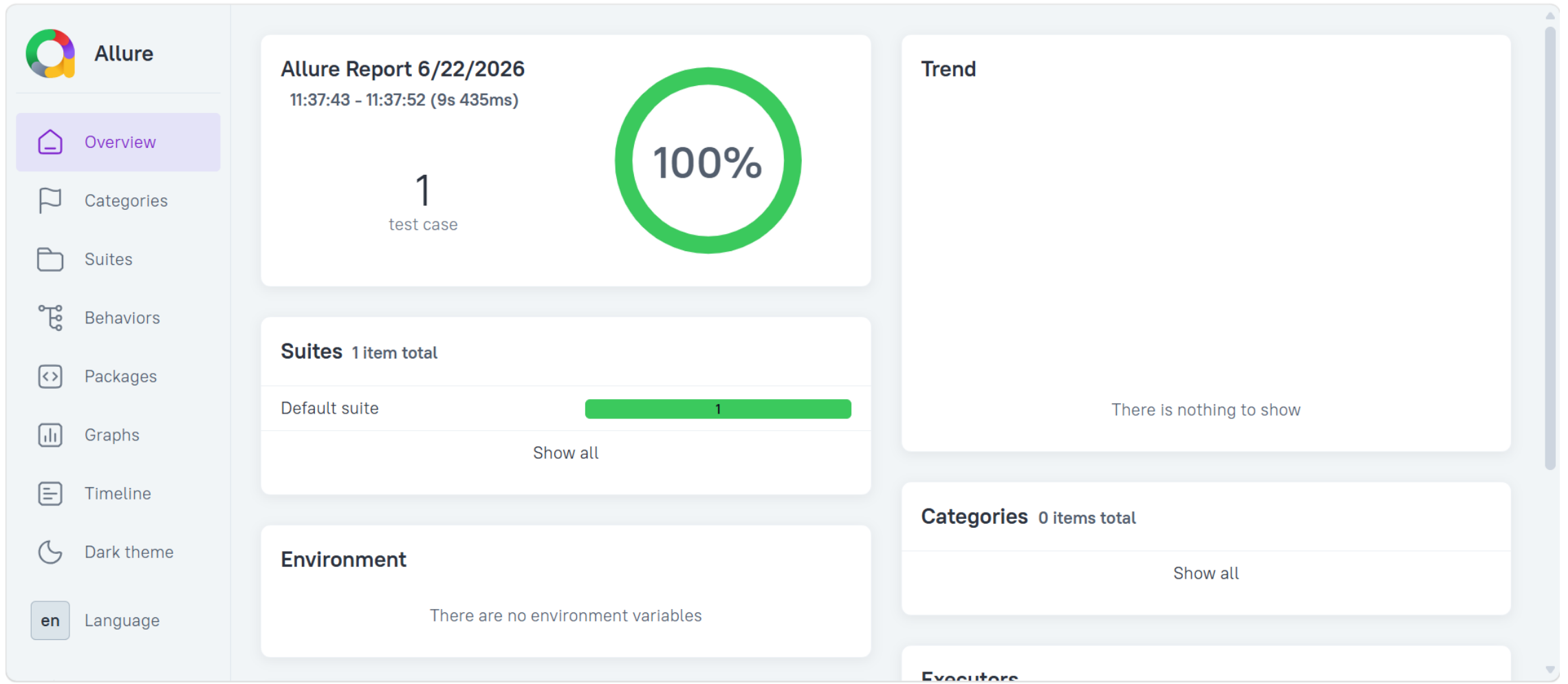
Run ALL
public class LoginIxigoTest extends BaseClass {
    @Test

    Run / Debug
    public void loginApp() throws Exception {

        ExcelDataProvider excel = new ExcelDataProvider();
        excel.getStringData("LoginIxigo", 0, 0);

        LoginPage loginHRM = PageFactory.initElements(driver, LoginPage.class);
        loginHRM.login_EasyCal(excel.getStringData("LoginIxigo", 0, 0), excel.getStringData("LoginIxigo", 0, 0));
        Thread.sleep(10000);
    }
}
```

Allure Report

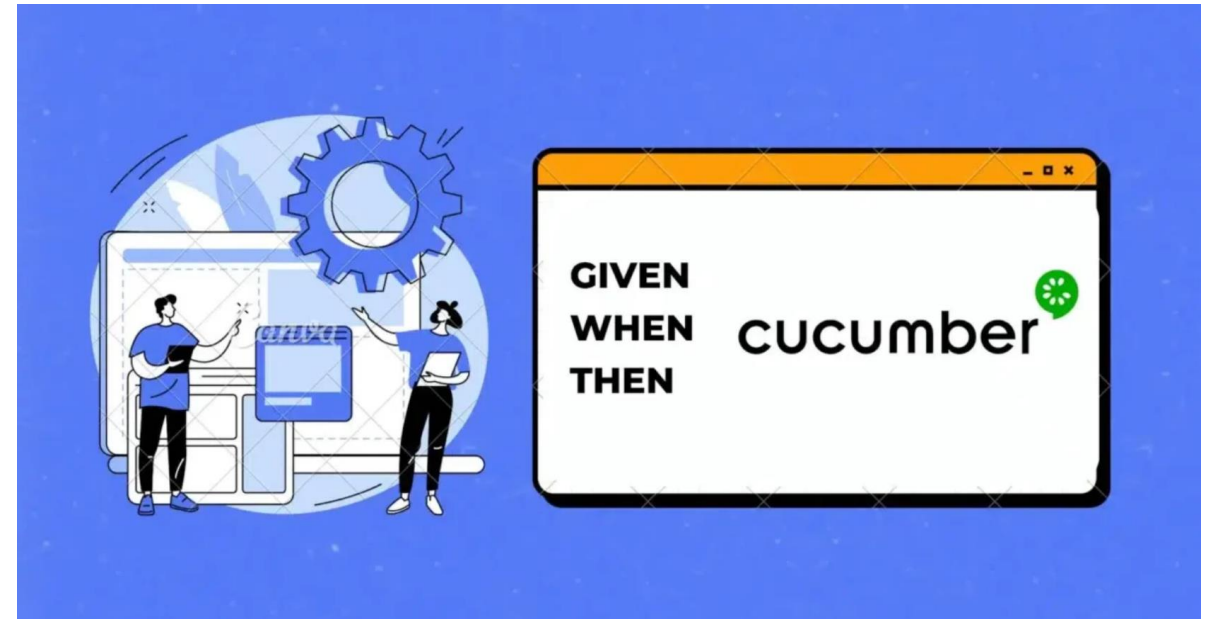


Cucumber

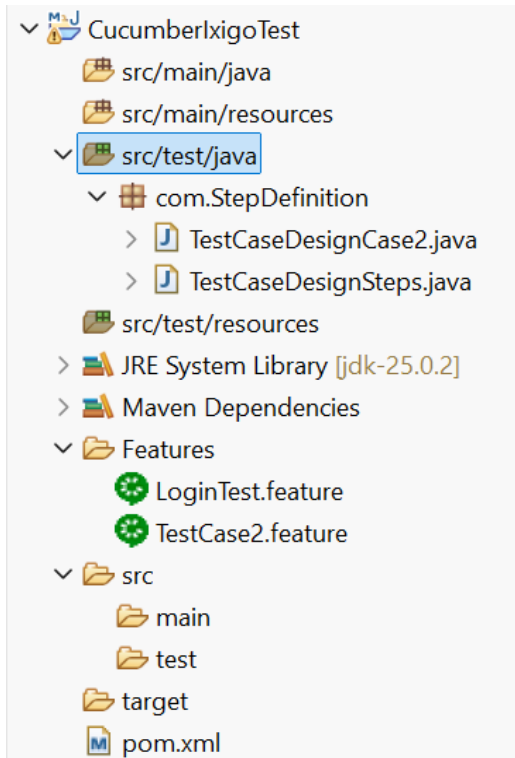
Cucumber is a testing tool used for **Behavior Driven Development (BDD)** that allows test cases to be written in simple, human-readable language.

It helps bridge the gap between **technical and non-technical stakeholders**.

- Uses **Gherkin language** (Given, When, Then)
- Improves collaboration between team members
- Connects test steps with automation code



Cucumber - Project Structure



Feature: Checking Login & Logout Functionality Run | Debug | Profile

Scenario: Successful Login with Valid Credentials Run | Debug | Profile

Given User should Open Chrome Browser
When User should Enter Url in Browser
When User Should Navigate Home Page
And Enter UserName and Password in Edit Box
And Click On Login PushButton
Then Message displayed Login Successfully

```
package com.StepDefinition;

import org.openqa.selenium.By;

public class TestCaseDesignSteps {
    WebDriver driver;

    @Given("User should Open Chrome Browser")
    public void user_should_open_chrome_browser() {
        driver= new ChromeDriver();
        driver.manage().window().maximize();
    }

    @When("User should Enter Url in Browser")
    public void user_should_enter_url_in_browser() {
        driver.get("https://www.ixigo.com/login");
    }

    @When("User Should Navigate Home Page")
    public void user_should_navigate_home_page() {
        String HomePage=driver.getTitle();
        System.out.println(HomePage);
    }

    @When("Enter UserName and Password in Edit Box")
    public void enter_user_name_and_password_in_edit_box() {
        driver.findElement(By.id("username")).sendKeys("student");
        driver.findElement(By.id("password")).sendKeys("Password123");
    }

    @When("Click On Login PushButton")
    public void click_on_login_push_button() {
        driver.findElement(By.xpath("//*[id=\"submit\"]")).click();
    }
}
```

API Testing

API Testing is the process of testing application Programming interfaces to ensure they function Correctly, securely and efficiently.

It focuses on validating the communication Between different software components by checking requests and responses without using the user Interface.



Apache JMeter

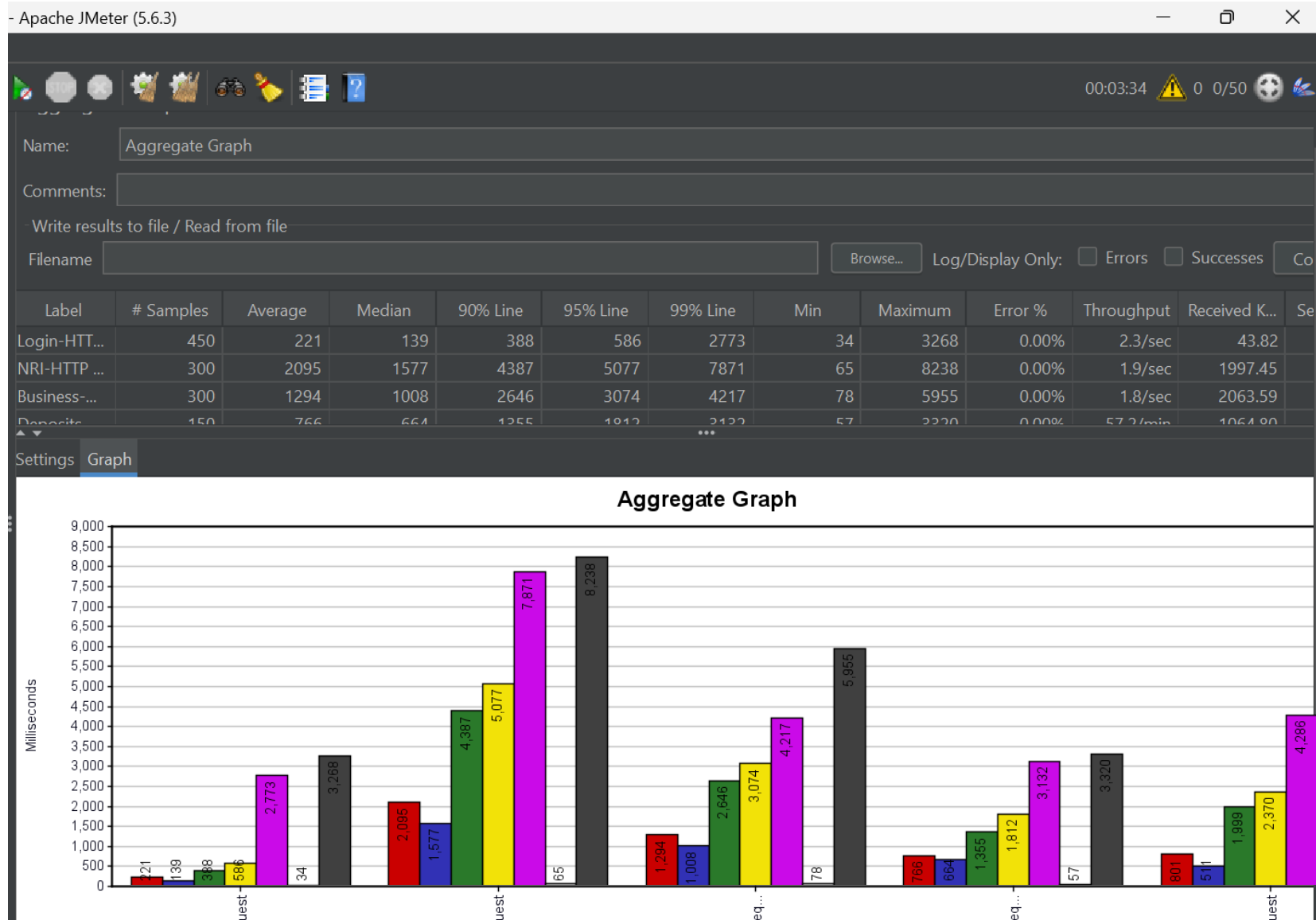
JMeter is an open-source tool used for **performance and load testing** of web applications and APIs.

It helps measure how a system performs under different levels of user load.

- Used for **load testing and performance testing**
- Simulates multiple users sending requests
- Measures response time and system behavior
- Helps identify bottlenecks



Aggregate Graph



JMeter Report

Apache JMeter Dashboard

Dashboard

Charts

Customs Graphs

Test and Report information

Source file	"test.csv"
Start Time	"6/21/26, 6:08 PM"
End Time	"6/21/26, 6:09 PM"
Filter for display	""

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.750	500 ms	1 sec 500 ms	Total
0.000	500 ms	1 sec 500 ms	Home HTTP Request
0.500	500 ms	1 sec 500 ms	Home HTTP Request-0
1.000	500 ms	1 sec 500 ms	Login HTTP Request
1.000	500 ms	1 sec 500 ms	Find Flights HTTP Request
1.000	500 ms	1 sec 500 ms	Register HTTP Request
1.000	500 ms	1 sec 500 ms	Home HTTP Request-1

Requests Summary



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label ▲	#Samples ⇅	FAIL ⇅	Error % ⇅	Average ⇅	Min ⇅	Max ⇅	Median ⇅	90th pct ⇅	95th pct ⇅	99th pct ⇅	Transactions/s ⇅	Received ⇅	Sent ⇅
Total	6	0	0.00%	840.17	436	1817	486.00	1817.00	1817.00	1817.00	0.16	0.86	0.02
Find Flights HTTP Request	1	0	0.00%	486.00	486	486	486.00	486.00	486.00	486.00	2.06	14.56	0.25
Home HTTP Request	1	0	0.00%	1817.00	1817	1817	1817.00	1817.00	1817.00	1817.00	0.55	3.90	0.13
Home HTTP Request-0	1	0	0.00%	1376.00	1376	1376	1376.00	1376.00	1376.00	1376.00	0.73	1.07	0.08
Home HTTP Request-1	1	0	0.00%	436.00	436	436	436.00	436.00	436.00	436.00	2.29	12.87	0.27
Login HTTP Request	1	0	0.00%	486.00	486	486	486.00	486.00	486.00	486.00	2.06	11.54	0.24
Register HTTP Request	1	0	0.00%	440.00	440	440	440.00	440.00	440.00	440.00	2.27	14.10	0.27

PostMan

The screenshot shows the Postman interface for a POST request named 'getrequestmethod'. The breadcrumb path is 'BOOTCAMP TESTING > API Testing > getrequestmethod'. The request method is 'POST' and the URL field contains the placeholder 'Enter URL or paste text'. The 'Send' button is visible. The 'Scripts' tab is active, showing a JavaScript test script with three assertions. The 'Test Results' section shows that all three assertions passed.

HTTP BOOTCAMP TESTING > API Testing > getrequestmethod Save Share

POST Enter URL or paste text Send

Docs Params Authorization Headers 7 Body **Scripts** Settings Cookies

Pre-req
Post-res ●

```
1 pm.test("Status code is 201", function () {  
2   pm.response.to.have.status(201);  
3 });  
4  
5 pm.test("Response body contains an 'id' field", function () {  
6   var jsonData = pm.response.json();  
7   pm.expect(jsonData).to.have.property("id");  
8 });  
9  
10 pm.test("The 'id' field in the response is a number", function () {
```

Test Results 🔄 201 Created • 734 ms • 1.36 KB • 🌐 | e.g. Save Response ⋮

Filter Results 🔄

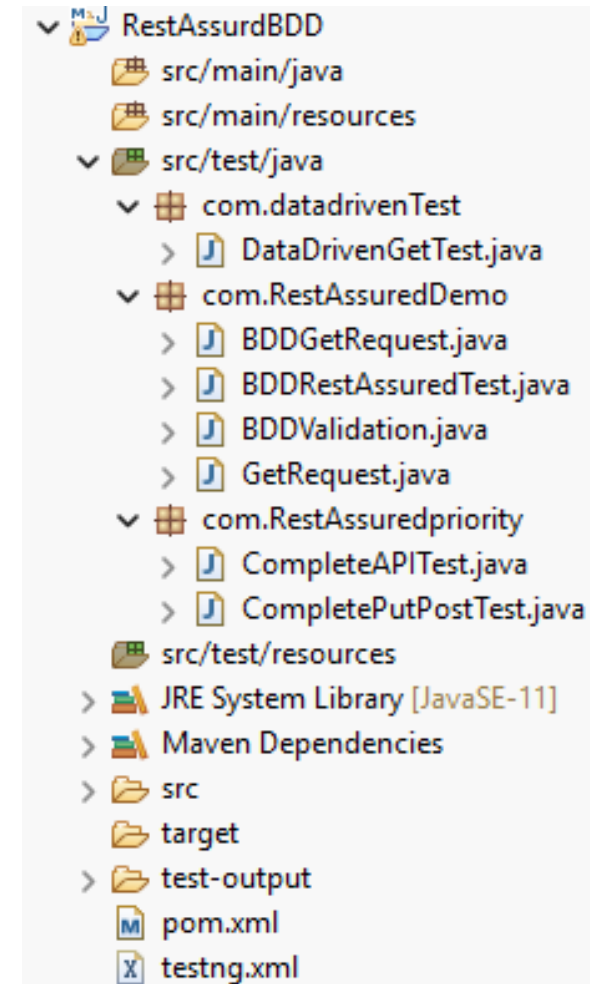
- PASSED Status code is 201
- PASSED Response body contains an 'id' field
- PASSED The 'id' field in the response is a number

Rest Assured

Rest Assured is a **Java-based library** used for automating **API testing** of RESTful web services.

It simplifies sending requests and validating responses in APIs.

- Supports methods like GET, POST, PUT, DELETE
- Easy integration with TestNG and Maven
- JSON/XML response validation
- Works well with automation frameworks



RestAssured – Sample Script

```
package com.RestAssuredDemoCaseStudy;
```

```
import org.json.JSONObject;  
import org.testng.annotations.Test;
```

```
import io.restassured.RestAssured;
```

```
import static io.restassured.RestAssured.*;  
import static org.hamcrest.Matchers.*;
```

Run All

```
public class PostRequestBDD {
```

```
    @Test
```

Run | Debug

```
    public void postRequestBDDTest() {
```

```
        RestAssured.baseURI = "https://jsonplaceholder.typicode.com";
```

```
        JSONObject request = new JSONObject();
```

```
        request.put("title", "Test Post");
```

```
        request.put("body", "Hello API");
```

```
        request.put("userId", 1);
```

```
        given()
```

```
            .log().all()
```

```
            .header("Content-Type", "application/json")
```

```
            .body(request.toString())
```

```
        .when()
```

```
            .post("/posts")
```

```
        .then()
```

```
            .log().all()
```

```
            .statusCode(201)
```

```
            .body("title", equalTo("Test Post"))
```

```
            .body("body", equalTo("Hello API"))
```

```
            .body("userId", equalTo(1))
```

```
            .body("id", notNullValue())
```

```
            .time(lessThan(5000L));
```

```
    }  
}
```

Git Repositories

Module Name: GitHub Repositories		Submitted By : Sahil Kumar Singh	
No.of Repositories: 21		Submitted To: Raghavendra BN	
Emp ID: 85009884			
S.No	Date	Project Name	Repository Link
1	03/06/2026 - 05/06/2026	Java Programs	https://github.com/SahilSingh77/Basic-Java-Programs
2	08/06/2026	SeleniumWebDriver-BrowserTesting	https://github.com/SahilSingh77/HYD-BootCamp-Testing-BrowserTesting
3	08/06/2026	TestNG-BrowserTesting, SeleniumWebDriver-CaseStudy1-BrowserTesting	https://github.com/SahilSingh77/HYD-BootCamp-Testing-BrowserTesting
4	09/06/2026	SeleniumWebDriver -CaseStudy -Commands,SeleniumWebDriver -CaseStudy_Locaters-EasyCalculation	https://github.com/SahilSingh77/HYD-BootCamp-eleniumWebdriver-Locators
5	10/06/2026	SelenuiumWebBrowser _Case Study-DropDownList,Radio Button	https://github.com/SahilSingh77/Selenium-DropDownBox_RadioButton_Navigation-Commands
6	10/06/2026	SeleniumWebDriver - AlertsTesting	https://github.com/SahilSingh77/Selenium-Alerts
7	10/06/2026	SeleniumWebDriver-DragAndDrop, ScrollUpAndDown-ActionTesting	https://github.com/SahilSingh77/Selenium-Actions
8	11/06/2026	SeleniumWebDriver_CaseStudy-POM-Practicetestautomation.,	https://github.com/SahilSingh77/HYD-BootCampTesting-Maven
9	11/06/2026	SeleniumWebDriver_CaseStudy-POM-Profilebrightest website	https://github.com/SahilSingh77/HYD-BootCampTesting-Maven
10	11/06/2026	SeleniumWebDriver_CaseStudy-POM- EasyCalculation Website,	https://github.com/SahilSingh77/HYD-BootCampTesting-Maven
11	11/06/2026	SeleniumWebDriver_CaseStudy-POM-Chase website	https://github.com/SahilSingh77/HYD-BootCampTesting-Maven
12	12/6/2026	DataDrivenTesting- ApachePOI	https://github.com/SahilSingh77/DataDrivenTestingUsingApachePOI
13	12/06/2026	Hybrid Driven Framework	https://github.com/SahilSingh77/Selenium-HybridDrivenFramework
14	13/06/2026	SeleniumWebDriver-DataProvider-TestNG	https://github.com/SahilSingh77/Selenium-DataProvider-TestNG
15	13/06/2026	Synchronization-Selenium	https://github.com/SahilSingh77/Selenium-Synchronization
16	11/06/2026	Selenium- ScreenShotTesting	https://github.com/SahilSingh77/Selenium-ScreenShotTesting
17	11/06/2026	DataDrivenTesting-Using CSV File	https://github.com/SahilSingh77/DataDrivenTesting-UsingCSVFile
18	13/06/2026	Parallel Testing - Using TestNG	https://github.com/SahilSingh77/ParallelTesting-UsingTestNG
19	16/06/2026	Assignment - SeleniumWebDriver	https://github.com/SahilSingh77/Assignment-CaseStudy-SeleniumWebDriver
20	17/06-2026	Cucumber - TestCases	https://github.com/SahilSingh77/BootCamp-Testing--CucumberAutomationTesting
21	19-06-2026	RestAssured - TestCases	https://github.com/SahilSingh77/RestAssured-TestCases



Thank You

Name: Sahil Kumar Singh

Emp Id : 85009884